VideoComposer: Compositional Video Synthesis with Motion Controllability

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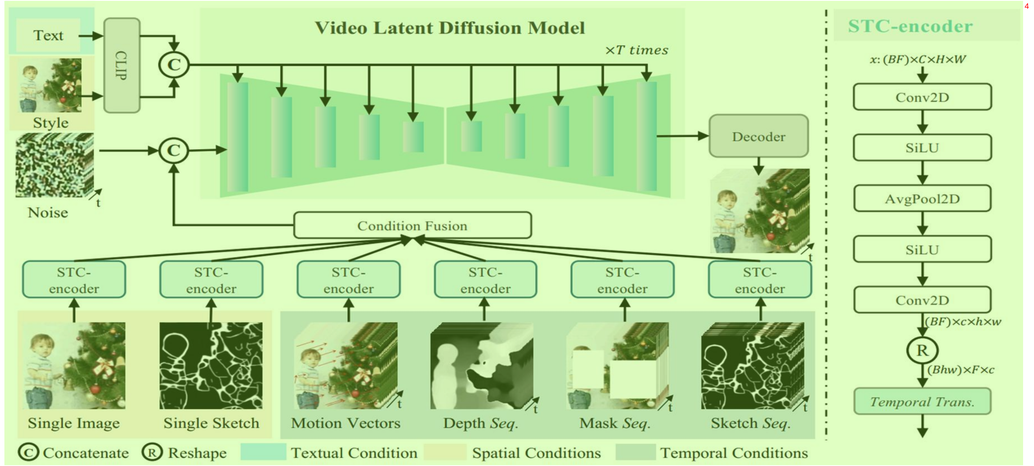
Introduce

- For the first time, we leverage motion vectors as a temporal condition, providing an efficient, compact, and robust solution for highly controllable video synthesis;
- VideoComposer allows users to flexibly compose a video with textual conditions, spatial conditions, and more importantly temporal conditions.

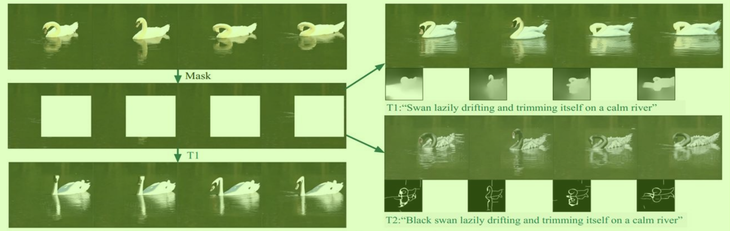
Framework of VideoComposer



Experimental Results

- **Compositional image-to-video generation**

- **Compositional video inpainting**



- **Compositional sketch-to-video generation**

- **Versatile motion control using hand-crafted motions**

- **Motion transfer**